Master Component Catalogue: Platform Super Admin Module (`ADM`)

Total **28 granular components** organised by the 6 side panel sections:

Section 1: Executive Intelligence (4 Components)

- `ADM-A01` : **Global Command Dashboard** — Pan-state rollup of live revenue, total registered parcels, active grievances, and citizen digital adoption.
- `ADM-A02` : **State & City Benchmarking Matrix** — Comparative league tables, DCB tax recovery percentages, grievance SLA compliance, and Swachh rankings.
- `ADM-A03` : **Consolidated Financial Ledger** — Central grants tracking (15th Finance Commission, AMRUT, SBM) and aggregated multi-tenant balance sheets.
- `ADM-A04` : **Spatial GIS Multi-City Viewer** — Cross-jurisdiction PostGIS map viewer displaying parcels, utilities, zones, and road networks across states.

Section 2: Tenancy & Hierarchy (4 Components)

- `ADM-T01` : **State Tenant Provisioning** — Onboard and configure Level 1 State entities (e.g., Telangana `TG` , Maharashtra `MH`), timezone, and official language packs.
- `ADM-T02` : **Sub-Tenant (ULB / City) Provisioning** — Create Level 2 child entities (e.g., GWMC, GHMC, CDMA-TP) with ward/zone hierarchies and administrative grades.
- `ADM-T03` : **Tenant Lifecycle & Status Controls** — Suspend, lock, archive, decommission, or put specific tenant workspaces into maintenance mode.
- `ADM-T04` : **White-Labelling & Custom Domain Router** — CNAME mapping, SSL certificate binding, official state emblems, and municipal branding watermarks.

Section 3: Modules & Subscriptions (4 Components)

- `ADM-S01` : **19-Module Master Catalogue** — Platform-wide repository of the 19 standard municipal modules and their 415 component definitions.
- `ADM-S02` : **State Module Subscription Allocator** — Assign, renew, or revoke module subscriptions for Level 1 State Tenants.
- `ADM-S03` : **Sub-Tenant Cascade Delegation Guard** — Rule engine ensuring child Sub-Tenants only receive module components authorized by their parent State.
- `ADM-S04` : **Quota & Usage Metering** — Per-tenant allocation and tracking of document storage caps (DigiLocker/S3), DLT SMS credits, and API rate limits.

Section 4: Global Identity & RBAC (5 Components)

- `ADM-R01` : **Global User Administration & Invitations** — Multi-tier user directory to provision, invite, and manage Level 0, Level 1, and Level 2 users.
- `ADM-R02` : **Cross-Tenant Scope & Context Review** — Audit and manage users holding multi-jurisdiction scopes (e.g., regional state directors overseeing multiple ULBs).
- `ADM-R03` : **415 Component RBAC Engine** — Granular permission matrix builder managing `{MODULE}-{ID}:{ACTION}` access rights and role templates.

- `ADM-R04` : **Support Impersonation Hub** — Secure, time-limited, cryptographically audited "Log in as Tenant Officer" facility for Level 3 support.
- `ADM-R05` : **Universal SSO & Keycloak Realms** — Keycloak identity broker managing MeriPehchan, Aadhaar e-Pramaan, SAML/OIDC federations, and MFA rules.

Section 5: Integrations & Compliance (5 Components)

- `ADM-C01` : **Central Gateway & Rate Limiting** — APISIX / Kong API gateway policies, per-tenant throttling, IP whitelisting, and WAF rules.
- `ADM-C02` : **Payment Switch & Escrow Hub** — Configure aggregator credentials (SBI ePay, BillDesk, Razorpay, BBPS) and route settlements to city treasury bank accounts.
- `ADM-C03` : **State Adapter Gateway** — Connectors for state-wide systems: IGRS land registration sync, eCourts CNR litigation sync, CGF/Treasury, and DISCOMs.
- `ADM-C04` : **CERT-In Immutable Audit Vault** — Append-only security audit log repository with cryptographic hash-chaining across all multi-tenant actions.
- `ADM-C05` : **Data Sovereignty & RLS Verification** — Continuous verification of PostgreSQL Row-Level Security policies to prevent cross-tenant data leakage.

Section 6: Platform Fleet & Operations (6 Components)

- `ADM-O01` : **Microservice Cluster Health** — Real-time telemetry, pod topology, async worker queues, and connection pool latency for all 19 module services.
- `ADM-O02` : **Multi-Tenant Schema Migrator** — Rolling, zero-downtime database DDL migration executor across all tenant partitions.
- `ADM-O03` : **Scoped Cache Partition Purge** — Targeted invalidation of tenant Redis cache keys (`stats:tenant:{id}:*`) without flushing global shared cache.
- `ADM-O04` : **Backup & Disaster Recovery** — Scheduled snapshot policies, Point-in-Time-Recovery (PITR) drill triggers, and cross-region failover tests.
- `ADM-O05` : **Global Broadcast & Alert Banner** — Platform-wide or state-scoped maintenance notifications, emergency alerts, and system broadcast banners.
- `ADM-O06` : **Version & Release Registry** — Track deployed software versions, manage feature flags, and schedule rollout rings (Canary, Blue/Green).

Total Super Admin Component Count: 28 Components